MATH 423: Lecture 7

Forward Contracts

Jiamin Jian

University of Michigan

Winter 2026

ONLY FOR MATH 423 WINTER 2026
NO PERMISSION FOR UPLOADING
HAS BEEN GIVEN

Contents

(1) Forward contracts
(2) Payoff graphs
(3) Forward contracts: no dividend
(4) Forward contracts: with discrete dividend

ONLY FOR MATH 423 WINTER 2026
NO PERMISSION FOR UPLOADING
HAS BEEN GIVEN

Forward Contracts

Definition 7.1

A forward contract is an agreement to buy or sell a risky asset (typically a stock) at a fixed future date T, called delivery date, for a price $F(0,T)$ determined today, called forward price.

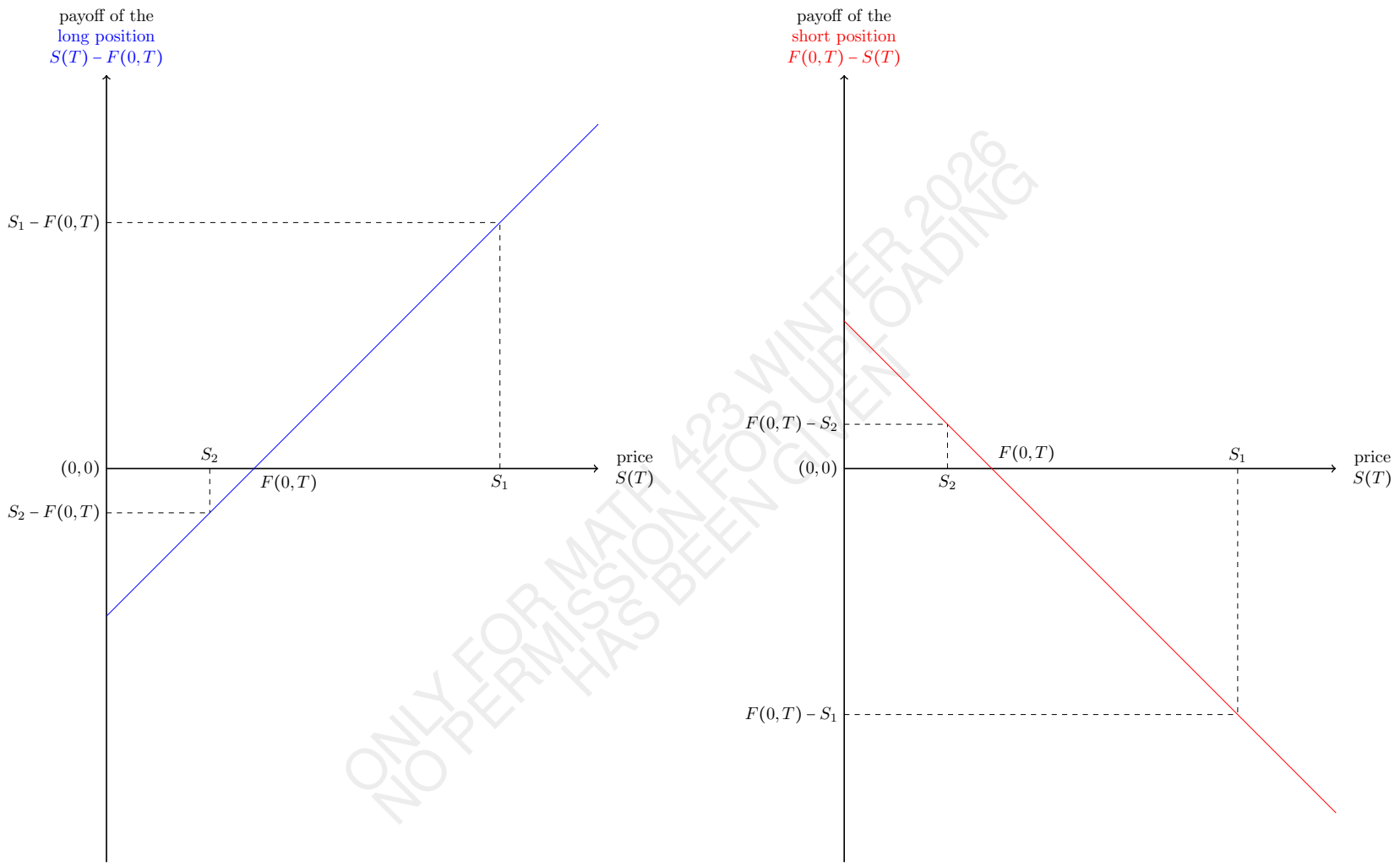
Remark 7.1

- The principal reason for entering into a forward contract is to become independent of the unknown future price of a risky asset.
- In the agreement, the party obliged to buy (sell) the asset is said to take a long (short) forward position.
- Forward contracts are designed to enter freely, i.e., No initial payment.
- The forward price is not the value of the forward contract.

Let $S(t)$ be the underlying price of the asset at time t. Then

- The future value of a long forward position at time T is $S(T) - F(0,T)$, which may be $+$ or $-$.
- The future value of a short forward position at time T is $F(0,T) - S(T)$, which may be $+$ or $-$.

Forward Contracts (payoff graphs)



ONLY FOR MATH 423 WINTER 2026
NO PERMISSION FOR UPLOADING
HAS BEEN GIVEN

Figure: Payoff for long and short Forward positions at the delivery day.

Forward Contracts (no dividend)

Recall: Prop 1.2 states that $F = S(0)\frac{A(1)}{A(0)}$.

Theorem 7.1

The no-arbitrage forward price is

$$F(0,T) = S(0)\frac{A(T)}{A(0)} = S(0)e^{rT},$$

where r is the continuous compounding interest rate.

Note: There is no $S(T)$ in the expression of $F(0,T)$. We cannot set the "fair" price $F(0,T) = S(T)$.

Why? Because the forward price $F(0,T)$ is determined today.

Proof: Similar to the proof of Prop 1.2 by using the no-arbitrage principal.

Exercise 7.1

If $F(0,T) < S(0)\frac{A(T)}{A(0)}$, then there is an arbitrage opportunity.

ONLY FOR MATH
NO PERMISSION
HAS BEEN

Remark 7.2

Some variations of $F(0,T) = S(0)\dfrac{A(T)}{A(0)} = S(0)e^{rT}$ in Thm 7.1:

- If interest is periodically compounded with frequency m and nominal interest rate r, then

$$F(0,T) = S(0)\frac{A(T)}{A(0)} = S(0)\Big(1+\frac{r}{m}\Big)^{mT};$$

- In terms of zero-coupon bond, no assumption about constant interest rates:

$$F(0,T) = S(0)\frac{A(T)}{A(0)} = S(0)\frac{1}{B(0,T)};$$

- If contract starts at time $t \in (0,T)$, we denote the corresponding forward price by $F(t,T)$. Then

$$F(t,T) = S(t)\frac{A(T)}{A(t)} = S(t)e^{r(T-t)}(> S(t)).$$

  Note that $\lim_{T\downarrow t} F(t,T) = S(t)$.

  If $S(t)$ is a continuous process with respect to the time t, then $\lim_{t\uparrow T} F(t,T) = S(T)$.

Example 7.1

Assume that $S(\frac{1}{4}) = 0.9S(0)$. If the continuously compounded annual interest rate is constant with $r = 6\%$, what is the percentage drop of the forward price at $t = \frac{1}{4}$ compared to $t = 0$ for a forward contract with delivery day $T = \frac{3}{4}$, that is, what is $\frac{F(\frac{1}{4},\frac{3}{4})-F(0,\frac{3}{4})}{F(0,\frac{3}{4})}$?

We need initially to determine $F(0, \frac{3}{4})$ and $F(\frac{1}{4}, \frac{3}{4})$. To this end,

$$F(0, \frac{3}{4}) = S(0)e^{0.06\times\frac{3}{4}} \text{ and } F(\frac{1}{4}, \frac{3}{4}) = 0.9 \times S(0)e^{0.06\times(\frac{3}{4}-\frac{1}{4})}.$$

Hence, the required percentage is

$$\frac{F(\frac{1}{4}, \frac{3}{4}) - F(0, \frac{3}{4})}{F(0, \frac{3}{4})} = \frac{0.9e^{0.06\times\frac{1}{2}} - e^{0.06\times\frac{3}{4}}}{e^{0.06\times\frac{3}{4}}} = -0.1134.$$

ONLY FOR MATH 420 WIN
NO PERMISSION FOR APPL
HAS BEEN GIVEN

Example 7.2

Suppose that $S(0) = 50,\ F(0,3) = 65,\ r = 10\%$ and short sale of stocks requires a 25% security deposit attracting interest at $d = 5\%$ (both rates are of continuously compounding). Is there an arbitrage opportunity? Find the highest rate d for which no A.O. exists.

Note: If short selling, the proceeds will split into two parts, 25% is used as security deposit which attracts interest at a lower rate $d = 5\%$, while the rest 75% can earn interest at the normal rate $r = 10\%$.

Check: $F(0,3) = 65$, $S(0)e^{rT} = 50 * e^{0.1*3} = 67.49$. There might be an A.O. if we buy a forward contract and short sell one share of the stock.

For such a portfolio,

$$V(T) = 25\%S(0)e^{dT} + 75\%S(0)e^{rT} - S(T) + [S(T) - F(0,3)] = 0.14.$$

So there is A.O.

In general, set $25\%S(0)e^{dT} + 75\%S(0)e^{rT} - S(T) + [S(T) - F(0,3)] \leq 0$. We get $d \leq 4.67\%$.

ONLY FOR MATH 425 WITH
NO PERMISSION FOR USE OR
HAS BEEN GIVEN

Forward Contracts (with dividend)

What happens when the underlying asset pays dividends?

Remark 7.3

We will understand the dividend as a sum of money paid to the holder of an asset. For example:

- the board of the company X decides to pay a dividend of $\$1$. This means that the market price of its traded stock is reduced by $\$1$ at the time the dividend is paid and each shareholder will receive $\$1$ for each share that they own.
- assume that in your bank account apart from $\$$ you also hold €. The risk-free rate for investments in $ is not the same as the one for €. The difference on the interest rates can be regarded as "continuously paid dividend".
- If the underlying asset is a commodity, storage costs may apply, *i.e.*, the holder of the underlying asset has to *pay* additional cash until the delivery day. The term holding cost describes the cash flow involved in maintaining a long or a short position in the underlying asset.

Forward Contracts (discrete dividend)

Theorem 7.2

The forward price of a stock paying dividend Div at time $t \in (0, T)$ is

$$F(0,T) = \left[S(0) - e^{-rt} Div\right] e^{rT} = \frac{S(0) - B(0,t)Div}{B(0,T)}.$$

This is more general than Thm 7.1: if no dividend, i.e. $Div = 0$, then we get Thm 7.1 exactly. Note that "Div" could be negative: e.g. holding cost.

Remark 7.4

The formula can easily be generalized to the case when dividends are paid more than once within $[0, T]$,

$$F(0,T) = \left[S(0) - Div_0\right] e^{rT} = S(0)e^{rT} - Div_T$$

where Div_0 (resp. Div_T) stands for present (resp. future) values of all dividends:

$$Div_0 = \sum_{i=1}^{n} d_i e^{-rt_i}, \quad Div_T = \sum_{i=1}^{n} d_i e^{r(T-t_i)}$$

given that dividend d_i is paid at time t_i.

Forward Contracts (discrete dividend)

Proof by no-arbitrage principle: Suppose that we start from 0 dollar.
If $F(0,T) > (<) \big[S(0) - e^{-rt}Div\big]e^{rT}$. Rule: Buy Low, Sell High. At time 0:

(a) The forward price $F(0,T)$ to trade the underlying asset is higher (lower) than the proposed one. So we take a short (long) forward position today in order to sell in a higher price (buy in a lower price);

(b) We do not want to be bothered by the random $S(T)$. So we buy one share today in order to sell it at time T (Since we will get a share at time T, we can short sell a share for $S(0)$ today);

(c) We borrow $S(0)$ from (deposit $S(0)$ in) the bank so that $V(0) = 0$;

At time t: we obtain a dividend in cash and deposit it instantly. (We have to pay the owner of the share a dividend at time t. To finance this payment, we have to borrow Div from the bank.)

At time T:

sell (buy) one share for $F(0,T)$ (and return to the owner)	$+(-)\ F(0,T)$
pay (get) $S(0)e^{rT}$ to the bank	$-(+)\ S(0)e^{rT}$
the deposit (loan) of Div at time t gives (needs)	$+(-)\ Dive^{r(T-t)}$

So the total wealth at time T is

$$V(T) = F(0,T) - S(0)e^{rT} + Dive^{r(T-t)} = F(0,T) - \big[S(0) - e^{-rt}Div\big]e^{rT} > 0$$

$$(-F(0,T) + \big[S(0) - e^{-rt}Div\big]e^{rT} > 0.)$$

COPY FOR MATH 233 WITH PERMISSION ONLY. NO PART MAY BE REPRODUCED OR DISTRIBUTED WITHOUT PERMISSION. HAS BEEN GIVEN

Exercise 7.3

$S(0) = 100$. Dividends of 3 dollars will be paid at $t_1 = \frac{1}{2}$ and 5 dollars will be paid at $t_2 = \frac{3}{2}$. The continuous interest rate is 10%. Forward price $F(0,2) = ?$

Solution: $Div_0 = 3e^{-0.1 \cdot 0.5} + 5e^{-0.1 \cdot 1.5} = 7.157$, then
$F(0,2) = \left[S(0) - Div_0\right] e^{rT} = (100 - 7.157)e^{0.1 \cdot 2} = 113.399$.

ONLY FOR MATH 423 WINTER 20
NO PERMISSION FOR UPLOADING
HAS BEEN GIVEN